

JAVA PROGRAMMING PROJECT REPORT

Hospital Management System

A Console-Based Application using Core Java, OOP & File Handling

Language: Java | Type: Console Application

Storage: File Handling (.txt)

Date: September 2026

1. Introduction

The **Hospital Management System** is a console-based Java application designed to simplify day-to-day hospital record-keeping. It replaces manual registers with a simple, menu-driven program that lets staff manage patients, doctors, appointments, and billing from one place.

The project applies core Object-Oriented Programming concepts along with Java's File I/O API to ensure that records persist even after the program is closed, without needing an external database.

2. Objectives

- Maintain accurate digital records of patients and doctors.
- Allow booking, viewing, and cancellation of appointments.
- Automatically generate a bill based on hospital stay and services.
- Demonstrate practical use of classes, collections, and file handling in Java.
- Provide a simple, distraction-free console interface usable without training.

3. System Requirements

Category	Requirement
Language	Java (JDK 17 or above)
IDE (optional)	VS Code / Eclipse / IntelliJ IDEA
Operating System	Windows / Linux / macOS
Storage	Local file system (.txt files)
RAM	1 GB minimum (lightweight console app)

4. System Design

The system follows a modular class design, separating data (entities) from logic (the driver class). This keeps the code organized and easy to extend.

Class	Responsibility
Patient	Stores patient details: ID, name, age, gender, disease, assigned doctor.
Doctor	Stores doctor details: ID, name, specialization, contact number.
Appointment	Links a patient and doctor with a date and status.
HospitalManagementSystem	Main class; handles the menu, business logic, and file I/O.

5. Modules Description

- **Patient Management** — Add, view, search, update, and discharge patient records.
- **Doctor Management** — Add, view, and search doctor profiles by ID.
- **Appointment Management** — Book new appointments, view all bookings, and cancel them.

- **Billing Module** — Calculates total charges from room rent, consultation fee, and medicine cost based on the number of days admitted.
- **File Handling Module** — Reads and writes all records to text files so no data is lost when the program restarts.

6. Sample Output

Below is a sample of the main menu as seen by the user when the program runs:

```
===== HOSPITAL MANAGEMENT SYSTEM =====  
1. Patient Management  
2. Doctor Management  
3. Appointment Management  
4. Generate Bill  
5. Save & Exit  
=====
```

7. Conclusion

The Hospital Management System successfully demonstrates how core Java concepts — classes, objects, collections, and file handling — can be combined to build a working, menu-driven application that solves a real-world problem. The project can be extended in future by integrating a database (JDBC/MySQL), adding a graphical interface using JavaFX or Swing, and introducing role-based login for admins, doctors, and staff.